

Name bhavya rana

Sapid 590034689

Batch 15

Algorithm

Main Algorithm

1. Start.
2. Display the menu:
 - Find the longest word in a sentence.
 - Check if one string is a rotation of another.
 - Exit.
3. Accept the user's choice.
4. Use a **switch** statement to perform the selected operation.
5. Call a separate user-defined function for each operation.
6. Repeat the menu until the user selects Exit.
7. Stop.

Operation 1 – Find the Longest Word

1. Read a sentence from the user.
2. Copy the sentence into a temporary string because `strtok()` modifies the string.
3. Split the sentence into words using `strtok()`.
4. Find the length of each word using `strlen()`.
5. Keep track of the word having the greatest length.
6. Display the longest word.
7. If two or more words have the same maximum length, the first one is displayed.

Operation 2 – Check String Rotation

1. Read two strings.
2. Compare their lengths.
3. If their lengths are different, they cannot be rotations.
4. Otherwise, concatenate the first string with itself.
5. Search for the second string inside the concatenated string using `strstr()`.
6. If found, the second string is a rotation of the first string.
7. Otherwise, it is not a rotation.

C Program

```
#include <stdio.h>
#include <string.h>

#define MAX 200
```

```

void findLongestWord(const char sentence[], char longest[]);
int isRotation(const char str1[], const char str2[]);

int main()
{
    int choice;
    char sentence[MAX];
    char longest[MAX];
    char str1[MAX], str2[MAX];

    do
    {
        printf("\n==== MENU =====\n");
        printf("1. Find the longest word in a sentence\n");
        printf("2. Check if one string is a rotation of
another\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(sentence, MAX, stdin);
                sentence[strcspn(sentence, "\n")] = '\0';

                findLongestWord(sentence, longest);

                printf("Longest word: %s\n", longest);
                break;

            case 2:
                printf("Enter first string: ");
                fgets(str1, MAX, stdin);
                str1[strcspn(str1, "\n")] = '\0';

                printf("Enter second string: ");
                fgets(str2, MAX, stdin);
                str2[strcspn(str2, "\n")] = '\0';

                if (isRotation(str1, str2))

```

```

        printf("The strings are rotations of each
other.\n");
    else
        printf("The strings are not rotations of
each other.\n");

    break;

case 3:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice. Please enter 1 to
3.\n");
}

} while (choice != 3);

return 0;
}

```

```

/* Function to find the longest word */
void findLongestWord(const char sentence[], char longest[])
{
    char temp[MAX];
    char *word;
    size_t maxLength = 0;

    strcpy(temp, sentence);
    longest[0] = '\0';

    word = strtok(temp, " ");

    while (word != NULL)
    {
        if (strlen(word) > maxLength)
        {
            maxLength = strlen(word);
            strcpy(longest, word);
        }

        word = strtok(NULL, " ");
    }
}

```

```
}
```

```
/* Function to check string rotation */
int isRotation(const char str1[], const char str2[])
{
    char combined[2 * MAX];

    if (strlen(str1) != strlen(str2))
        return 0;

    strcpy(combined, str1);
    strcat(combined, str1);

    if (strstr(combined, str2) != NULL)
        return 1;

    return 0;
}
```

Function Prototypes

```
void findLongestWord(const char sentence[], char longest[]);
int isRotation(const char str1[], const char str2[]);
```

Test-Case Table

Test	Operation	Input	Expected Output
1	Longest word	C programming is interesting	programming
2	Longest word	I love computers	computers
3	Longest word	Hello world	Hello
4	Longest word	Programming	Programmin g
5	Rotation	ABCD, CDAB	Rotation
6	Rotation	ABCD, DABC	Rotation
7	Rotation	ABCD, ACBD	Not rotation
8	Rotation	hello, llohe	Rotation
9	Rotation	hello, world	Not rotation
10	Invalid menu choice		5 Invalid choice
11	Exit		3 Exiting program

Output

```
Last login: Sat Oct 10 19:17:08 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day43.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day43.c -o day43
[bhavya@Bhavyas-MacBook-Air ~ % ./day43

===== MENU =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 1
Enter a sentence: C programming is intresting
Longest word: programming

===== MENU =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 2
Enter first string: ABCD
Enter second string: CDAB
The strings are rotations of each other.

===== MENU =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 3
Exiting program...
bhavya@Bhavyas-MacBook-Air ~ %
```